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REVIEW ARTICLE

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AI support in self-regulated learning: A decade of technological evolution and meta-analysis

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Abstract

This meta-analysis systematically examines 35 empirical studies (2013–2025) investigating artificial intelligence applications within Zimmerman's cyclical model of self-regulated learning (SRL). Three principal discoveries emerge: (1) Technological progression has evolved through three co-existing paradigms: rule-based architectures, data-driven adaptive systems and generative AI ecosystems, that demonstrate increasingly sophisticated capabilities for human-AI collaboration. (2) While AI-supported SRL interventions yield a moderate overall effect size ($g=0.507$), their impact is uneven; AI is significantly more effective during the task performance phase ($g=0.574$) than in the preparatory forethought phase ($g=0.401$). Notably, generative AI shows markedly superior efficacy across all phases (e.g. $g=0.709$ for forethought, $g=0.938$ for performance), though high heterogeneity suggests these effects are heavily contingent on specific instructional designs. (3) Moderator analysis identifies optimal contexts in secondary education, natural science disciplines, fully online settings and interventions of medium duration (2–10 weeks), while also revealing that effects are substantially larger when measured by behavioural traces compared to self-reports. Critically, these findings highlight a persistent performance-competence divide, suggesting that AI's capacity to scaffold immediate task performance may outpace its current ability to cultivate durable, transferable self-regulatory competence. The study discusses the implications of

Jun Xu and Yuying Luo contributed equally to this work and share first authorship.

this divide and proposes a research agenda focused on designing AI systems that foster genuine learner autonomy.

KEYWORDS

artificial intelligence, educational technology, large language models, meta-analysis, self-regulated learning

Practitioner notes

What is already known about this topic

- AI technologies have shown promise in supporting self-regulated learning (SRL) through personalized feedback and metacognitive scaffolding.
- Previous studies report inconsistent outcomes of AI interventions across SRL phases (forethought, performance, self-reflection), with limited exploration of phase-specific impacts
- Existing meta-analyses often treat SRL as a unified construct rather than examining how different AI types support distinct self-regulatory processes.

What this paper adds

- Demonstrates moderate overall effectiveness of AI-supported SRL ($g=0.507$) with differential impacts across phases: strongest during task performance ($g=0.574$), weaker in forethought ($g=0.401$) and self-reflection ($g=0.464$)
- Maps three coexisting AI paradigms—rule-based, data-driven and generative AI—revealing that generative AI achieves superior outcomes across all SRL phases, though effectiveness depends on alignment between AI affordances and specific self-regulatory processes.
- Identifies optimal implementation contexts through moderator analysis: secondary education settings, natural science disciplines, fully online environments and medium-duration interventions (2–10 weeks) yield stronger effects.
- Articulates a critical performance-competence divide, substantiated by quantifying how AI's impact on observable behaviours ($g=0.751$) is more than double its effect on learners' self-reported perceptions ($g=0.369$).

Implications for practice and/or policy

- Integrate AI paradigms strategically rather than viewing them as replacements: combine rule-based systems' stable scaffolding with generative AI's adaptive dialogue to support the full SRL cycle.
- Design interventions of medium duration (2–10 weeks) to optimize skill acquisition while maintaining engagement, avoiding both novelty effects and scaffold dependency.
- Adapt AI implementation to disciplinary demands: structured procedural support works well in natural sciences, while social sciences require enhanced support for open-ended inquiry and critical analysis.
- Develop assessment frameworks that measure delayed, unsupported performance alongside immediate gains to ensure AI fosters genuine self-regulatory competence rather than temporary performance enhancement.

INTRODUCTION

The rapid advancement of artificial intelligence (AI) and its growing impact on the workforce has intensified the need for lifelong learning, positioning self-regulated learning (SRL) as an essential competency for navigating societal and technological change (de Mooij et al., 2025). Supporting learners to become effective self-regulators remains a persistent pedagogical challenge, largely because SRL processes are complex, cyclical and often covert (Winne & Hadwin, 1998). AI technologies offer a potential solution by providing novel data-driven methods to make these internal learning processes more visible and supportable (Sposato, 2025). However, in traditional classroom settings, successful SRL often relies on learners' intrinsic motivation and teachers' individualized guidance (de Ruig et al., 2023; Li et al., 2024). As the demand for independent student learning increases and resources become relatively limited—particularly in online environments—maintaining effective SRL becomes increasingly difficult without timely interaction and emotional support (Jin et al., 2023).

Against this backdrop, AI technologies have demonstrated considerable promise for supporting SRL by addressing the scalability limitations of human scaffolding. These capabilities address many of the challenges identified in traditional learning environments, offering learners immediate assistance with goal setting, metacognitive monitoring and reflective practices (Chang & Sun, 2024; Rawas, 2024). However, research examining the effectiveness of AI interventions across the forethought, performance and self-reflection phases has yielded inconsistent findings. Current empirical studies typically focus on specific AI tools or individual SRL phases, lacking systematic quantitative evaluation across the complete regulatory cycle (Wu, 2024). This fragmented research landscape has resulted in a lack of integrated evidence, constraining the development of a comprehensive understanding of how AI's evolving affordances differentially support SRL processes.

To address these gaps, this study undertakes a systematic review of intervention-based research on AI-supported SRL over the past decade, employing the meta-analytic framework proposed by Pigott and Polanin (2020). By synthesizing empirical evidence, our goal was to trace the developmental trajectory of AI in SRL, evaluate its overall effectiveness and investigate variations in effect sizes across different paradigms of technological evolution and SRL processes. While recognizing that SRL unfolds recursively in practice, this phase-based analysis serves as a necessary heuristic to identify where AI support is most effective and where gaps remain. Specifically, the following research questions guide our inquiry:

RQ1. How has the trajectory of AI development in supporting SRL evolved over the past decade?

RQ2. What are the effect sizes of AI interventions across the forethought, performance and self-reflection phases of SRL?

RQ3. From the perspective of technological evolution, do different types of AI exhibit varying effect sizes in supporting SRL?

RQ4. What potential moderating variables significantly influence the intervention effectiveness of AI in SRL overall?

THEORETICAL FRAMEWORK AND PRIOR RESEARCH

Development of self-regulated learning theory and the focus of this study

The concept of SRL emphasizes learners' proactive management of cognition, motivation and behaviour to achieve specific learning goals (Zimmerman, 1989). Since the 1980s, SRL theory has progressed from an individualized, linear perspective to more dynamic, socially contextualized approaches. Bandura's (1991) social cognitive theory underscores the interplay between environment and self-efficacy, Pintrich's (2000) four-phase model highlights the essential role of motivation, Winne and Hadwin (1998) explore SRL from an information-processing and metacognitive standpoint, and Hadwin et al.'s (2017) socially shared regulation of learning (SSRL) framework focuses on self-regulation in collaborative group settings. Despite differing emphases, all the aforementioned theoretical frameworks converge on the notion that phased self-regulation can enhance academic performance.

In technology-enhanced learning, Zimmerman's three-phase model (forethought, performance and self-reflection) remains a widely accepted operational framework (Wong et al., 2019). As AI evolves into a learning partner, it introduces more complex, co-regulatory dynamics; however, the efficacy of such human–AI partnerships is predicated on the individual's core self-regulatory competencies, as effective shared regulation is built upon the skills of its individual members (Sharma et al., 2024). Zimmerman's (2002) cyclical model provides an essential, process-oriented lens to systematically map varied interventions onto the core components of individual learning. This framework's parsimonious structure and widespread adoption in AI-SRL research (Lee et al., 2025) enable the methodological consistency required for meta-analytic synthesis across diverse technological paradigms.

Compared to traditional contexts where SRL depends largely on learners' self-awareness or teacher guidance, AI technologies offer innovative means to assist SRL processes. Drawing on data from the Web of Science database retrieved in June 2025, Figure 1 shows the annual publication volume and citation trends related to SRL. This trajectory highlights significant growth periods, including an initial rise around 2007, a broadening of research scope around 2014 and a pronounced surge peaking in 2023. In the early 21st century, the

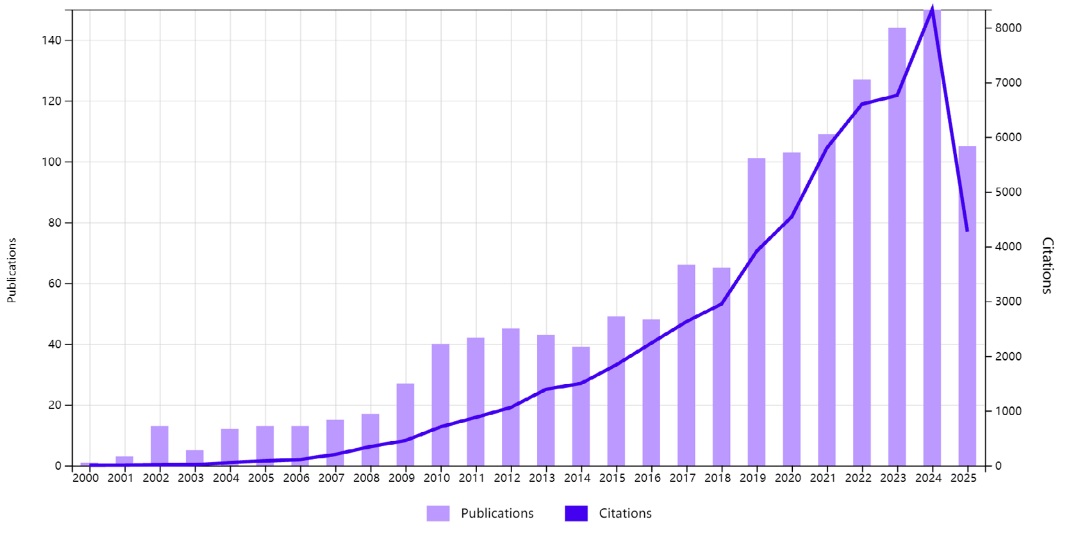

FIGURE 1 Annual publications and citation trends in self-regulated learning since the 21st century.

proliferation of internet and mobile devices enhanced access to digital learning resources (Dabbagh & Kitsantas, 2013). Subsequently, learning analytics and social media data broadened the reach of SRL research, while the recent emergence of generative AI has enabled highly personalized, real-time support, spurring a growing interest in using emerging technologies to strengthen SRL (Gao et al., 2024). As a result, SRL research has shifted from predominantly conceptual exploration to more data-driven empirical validation (Molenaar et al., 2023).

Meta-analytic research on technology-supported SRL: Progress and limitations

In recent years, numerous studies have examined the impact of various technologies on SRL and summarized their findings through meta-analyses. Overall, this body of work affirms SRL's crucial role in improving academic performance and supporting learner autonomy (Dent & Koenka, 2016; Dignath & Büttner, 2018). Researchers have further noted AI's potential to simulate teacher guidance, personalize strategy application and perform advanced learning analytics (Akamatsu et al., 2019; Dahri et al., 2024).

However, few studies have systematically focused on SRL as a skill to be cultivated, particularly with regard to emerging AI paradigms (Luckin et al., 2022). Existing meta-analyses often concentrate on specific technologies (Chen et al., 2024), leaving AI's distinct role in different SRL phases underexamined (Lu & Lin, 2024). Additionally, most research treats SRL as a unified construct rather than investigating how AI can support its forethought, performance and self-reflection phases individually. Drawing on Zimmerman's three-phase SRL model, this study quantitatively explores AI's impact on each phase, taking into account the varying developmental paradigms of AI technologies. By doing so, it addresses current gaps in meta-analytic SRL research and provides insights into how AI's evolving capabilities can enhance SRL across diverse educational contexts.

METHODS

This study adopts Zimmerman's SRL model as its theoretical basis and conducts a meta-analysis of empirical research published between 2013 and 2025 that applied AI technologies to SRL interventions. In line with the meta-analytic procedures outlined by Pigott and Polanin (2020), the research process involved (1) systematically searching and screening the literature, (2) coding and evaluating the quality of the selected studies, (3) calculating effect sizes using appropriate statistical methods and (4) interpreting the results comprehensively.

Given that AI-supported SRL spans a wide range of educational contexts and technological paradigms, maintaining high methodological standards is essential to minimize potential biases. To ensure research quality while maximizing inclusiveness, this study included peer-reviewed journal articles indexed in SSCI, SCIE, Scopus or EI, as well as conference proceedings from Scopus and EI databases that met rigorous quality criteria. All included studies underwent systematic quality assessment using the Medical Education Research Study Quality Instrument (MERSQI), which has been established as a reliable tool for evaluating methodological rigour in technology-enhanced learning reviews (Tian & Zheng, 2024). Following evidence that abstracts with MERSQI scores below 10 have significantly lower publication rates (Sawatsky et al., 2015), we adopted this score as our minimum threshold to ensure a methodologically robust evidence base.

Literature search and inclusion criteria

Adhering to the PRISMA guidelines, a systematic search was conducted in four major databases (ERIC, Ei Compendex, Scopus and Web of Science) to ensure thoroughness and quality (Xu et al., 2025). The search focused on SRL-related studies, with the time frame set from 2013 to 2025 to capture recent developments. For instance, in Web of Science, the search query was:

TS = ('self-regulat*' AND learn*) AND PY = 2013-2025. This query included various forms of 'self-regulation' and 'self-regulated', without strictly limiting the term 'AI', thereby encompassing diverse AI applications. Manual screening followed to identify studies that explicitly used AI technologies—such as intelligent tutoring systems, adaptive learning platforms or conversational AI tools—to support SRL.

Based on the above search strategy and the study's aims, the following inclusion criteria were established: (1) The study examined AI-supported SRL interventions, addressing at least one of the three SRL phases (forethought, performance or self-reflection). (2) It used experimental or quasi-experimental designs, comparing outcomes between an experimental group and a control group. (3) The intervention's effect was assessed using measures of SRL processes. Outcomes such as academic achievement were included only when theoretically justified as a direct consequence of the supported regulatory processes. To clarify the distinction between subject-matter and process-focused support, interventions were included only if their feedback was designed to function as an SRL scaffold—that is, to prompt regulatory processes such as self-evaluation and strategy adjustment, rather than merely to deliver content corrections. (4) It offered sufficient data (e.g. means, standard deviations, sample sizes) to calculate effect sizes. (5) It achieved a MERSQI score of 10 or higher, indicating acceptable methodological quality.

Following the initial screening, quality assessment was conducted on all potential studies using MERSQI, which evaluates six domains: study design, sampling, type of data, validity of evaluation instrument, data analysis and outcomes. Two studies with MERSQI scores below 10 were excluded due to methodological limitations. The final sample comprised 35 articles (mean MERSQI = 13.07, SD = 1.27), including 31 journal articles and 4 conference proceedings, all meeting the established criteria for quality and comparability. The complete list of included studies is provided in Appendix A. As shown in Figure 2, the final screening yielded 35 articles meeting the criteria for quality and comparability.

Study characteristics and intervention coding

To enable precise data extraction and facilitate comparison across studies, a systematic coding scheme was employed for study characteristics, AI interventions and SRL outcome measures (Table 1). First, participant details (e.g. geographical region, educational level, subject area and learning environment) were recorded to capture the context of each study. Additionally, recognizing the critical distinction between learners' perceptions of their self-regulation and actual regulatory behaviours (Winne, 2022), we coded each outcome's measurement type as either self-report or behavioural/trace. Next, AI interventions were classified according to their developmental paradigms and specific types (e.g. intelligent tutoring systems, adaptive learning platforms, conversational AI, intelligent feedback tools), and each intervention's targeted SRL phase was identified.

To investigate AI's impact on distinct SRL processes, relevant variables—such as goal setting, strategy planning, self-efficacy, metacognitive monitoring, time management and self-evaluation—were aligned with the appropriate SRL phases (Table 2),

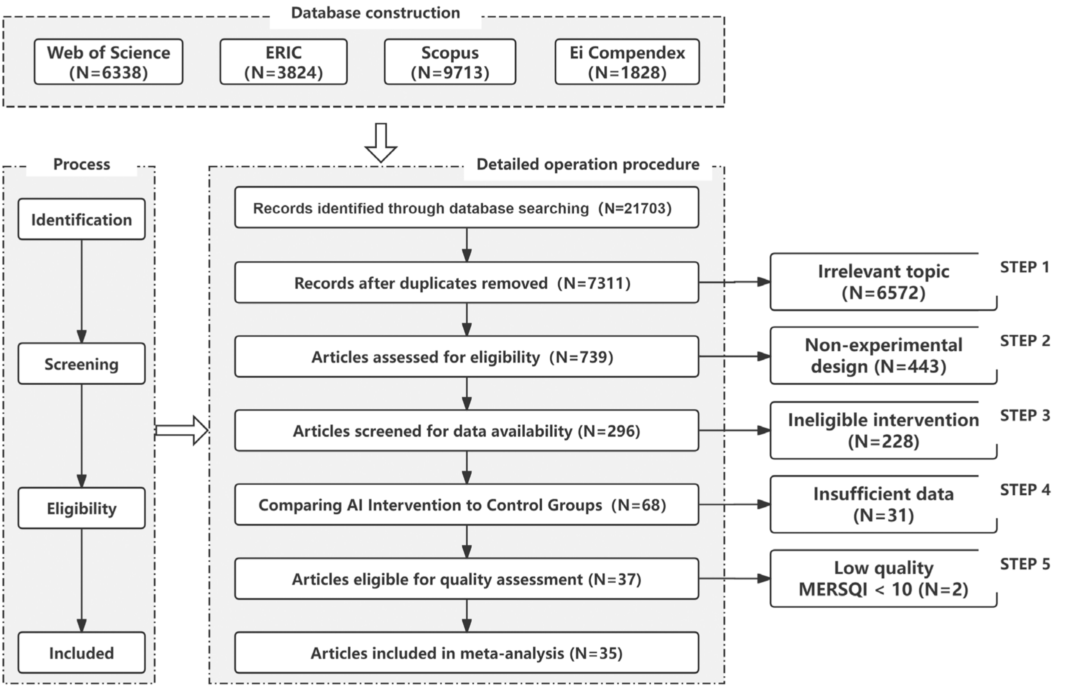

FIGURE 2 Search flow diagram.

with each distinct outcome treated as an independent effect size, ultimately yielding 133 effect sizes from the 35 included studies. While acknowledging that SRL processes are recursive and overlapping in authentic practice, this phased analysis is employed as an essential analytical heuristic. For a meta-synthesis of diverse literature, such a framework is methodologically necessary to systematically categorize disparate outcome measures and distil meaningful patterns from heterogeneous evidence (e.g. Hemmler & Ifenthaler, 2024). The classification of each variable was determined by its conceptual definition and its measurement point within a study's design (e.g. pre-task for planning, post-task for reflection), ensuring alignment with Zimmerman's (2002) cyclical model. To enhance analytical precision and directly address the critical distinction between perceived and actual regulation, we further coded each outcome measure by assessment type, distinguishing between self-report instruments (68% of effect sizes) that capture learners' perceptions and beliefs about their SRL, and behavioural/trace measures (32%) that document observable regulatory actions (Winne, 2022). This coding enabled a more transparent interpretation of whether the observed effects reflect changes in metacognitive awareness or actual behavioural manifestations of self-regulation.

In addition, to track how AI-supported SRL has evolved over the past decade, a longitudinal cluster analysis was conducted based on technological features, publication timelines and research methods (see Section 'A preliminary examination of AI development in the field of self-regulated learning (for RQ1)'). This analysis grouped AI tools into different stages of development. To ensure coding accuracy, two researchers independently coded 20 of the selected articles. Following Landis and Koch's (1977) guidelines, Cohen's kappa coefficients of 0.88 and 0.85 indicate strong interrater reliability; discrepancies were resolved through discussion, yielding a final coding scheme that is both accurate and reliable.

TABLE 1 Lists of codes for the analysis of selected articles.

Dimension	Item	Description
Participant characteristics	Region	Text records, for example, the United States, China, Japan, etc.
	Sample size	(1) 1–50; (2) 51–100; (3) 101–300.
	Educational level	(1) Primary school; (2) Junior and senior high school; (3) Institutions of higher education.
	Discipline	(1) Natural Science (including science, mathematics, physics, biology, geography); (2) Social Science (including politics, education, psychology, linguistics).
Content characteristics	Learning environment	(1) Online learning settings; (2) Blended learning settings.
	AI intervention method	A system that uses computational models (e.g. rule-based logic, machine learning, generative modeling) to provide adaptive or personalized feedback and scaffolding for SRL.
	SRL variables	Variables related to SRL recorded in the literature, such as self-efficacy, metacognition, attention focus and time management.

RESULTS

A preliminary examination of AI development in the field of self-regulated learning (for RQ1)

Analysis of the 35 studies included in this research reveals a technological trajectory defined by three co-existing paradigms. This evolution does not represent a linear displacement of earlier technologies but rather an expansion of the pedagogical landscape, progressing from structured, rule-based guidance to data-driven personalization and, most recently, to dialogic co-regulation with generative AI. Figure 3 illustrates the distribution of the 133 effect sizes from these studies across these AI paradigms and chronological periods, visually confirming that established rule-based and data-driven systems persist alongside emerging generative models rather than being displaced.

Paradigm 1: Rule-based intelligent tutoring systems

Foundational to the field and prominent since 2015, rule-based systems, such as MetaTutor (Duffy & Azevedo, 2015) and early intelligent tutoring systems (Mohammadzadeh & Sarkhosh, 2018) mainly depended on predefined rules and expert knowledge bases to provide structured but relatively rigid support. These systems helped learners with goal setting and strategy planning by offering metacognitive scaffolding and basic feedback (Jones et al., 2018). Although their capacity for contextual adaptation and deeper self-reflection was limited, they established a critical foundation for the broader, more versatile applications of subsequent AI technologies. Notably, such rule-based approaches have persisted in development and application, as evidenced by their continued presence in studies from 2021 to 2023 (Figure 3) and their ongoing utility in structured environments or for specific tasks where clarity and stability are paramount (Chu et al., 2023; Liang et al., 2024).

TABLE 2 Mapping of SRL indicators to phases.

Phase	Theoretical definition	Indicator	Definition
Forethought (pre-task)	The preparatory phase involving task analysis and mobilizing motivational beliefs	Goal setting and strategy planning	Establishing clear learning goals and implementation paths
		Self-efficacy	Confidence in one's ability to complete tasks
		Task value and interest	Awareness of the significance and enjoyment of learning tasks
		Outcome expectation	Predictions of the effort and potential benefits of learning
Performance (during task)	The execution phase involving self-control (e.g. strategy use) and self-observation	Task strategies	Applying cognitive and metacognitive skills (e.g. summarizing, monitoring)
		Attention focus and time management	Focusing on tasks and allocating study time effectively
		Self-observation and monitoring	Continuously tracking one's performance and strategy usage
		Seeking help	Identifying difficulties and actively seeking external assistance
Self-reflection (post-task)	The evaluative phase involving self-judgement of performance and adaptive self-reactions	Self-evaluation	Comparing outcomes with standards or initial expectations
		Causal attribution	Interpreting successes or failures as a basis for strategy modification or effort adjustment
		Adaptive inference and response	Adjusting future study behaviour based on reflective results
Comprehensive phase	An assessment of the overall effectiveness of an SRL-driven task, using measures not attributable to a single phase	Academic achievement	Test results or indicators of academic performance
		Skill acquisition and transfer	Application of skills and knowledge in new contexts
		Learner satisfaction and engagement	Emotional responses and investment in the learning process

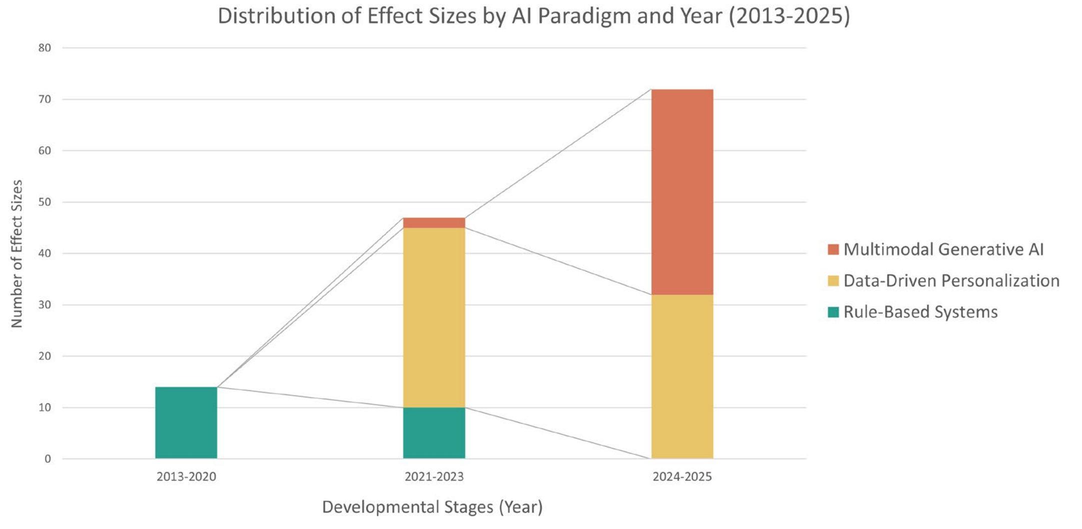

FIGURE 3 Distribution of effect sizes by AI paradigm and year ($k=133$ effect sizes).

Paradigm 2: The advancement of data-driven personalization

With the introduction of educational data mining, natural language processing (NLP) and complex system frameworks, AI tools expanded into a data-driven paradigm, emerging prominently from around 2021 and showing sustained presence (see Figure 3). These systems are distinguished by their capacity to learn from learner data to tailor support. More advanced learning analytics systems, such as ALEKS and Studybuddy, utilized real-time data to analyse learners' knowledge states and dynamically adjust learning paths and resources, guiding effective learning behaviours (Harati et al., 2021; Mejeh et al., 2024). Meanwhile, advancements in NLP and speech recognition enabled these tools to offer more flexible and personalized language instruction and instant feedback, significantly enhancing SRL's adaptability and contextual awareness (Chang et al., 2022; Wei, 2023). Additionally, the integration of complex systems theory provided researchers with new perspectives to understand learners' regulatory strategies and implement targeted interventions from a macro viewpoint (Hilpert et al., 2023). However, despite progress in personalization and adaptability, these tools still relied on pre-defined data and strategies, limiting the depth and creativity of interactions.

Paradigm 3: The integration of multimodal generative AI in intelligent educational ecosystems

Representing the most recent expansion of the SRL ecosystem, generative AI gained significant traction from 2023 onwards and is prominent in the 2024–2025 period within our sample (see Figure 3), AI tools incorporating large language models (LLMs), such as ChatGPT-Assisted Learning Aid and SRLbot, began to act as proactive cognitive partners, supporting student self-learning. By combining contextual prompts with personalized feedback, these tools facilitated autonomous problem-solving and higher order thinking skills (Ng et al., 2024). At the same time, virtual reality systems and multimodal feedback technologies that use image recognition and voice-based interactions contributed to multi-sensory learning experiences. These environments supported emotional regulation, encouraged creativity and promoted cyclical co-regulation processes (Hsu et al., 2023; Wang et al., 2024). Compared with earlier stages, these generative AI platforms focused on individual learners during instructional sessions, integrating

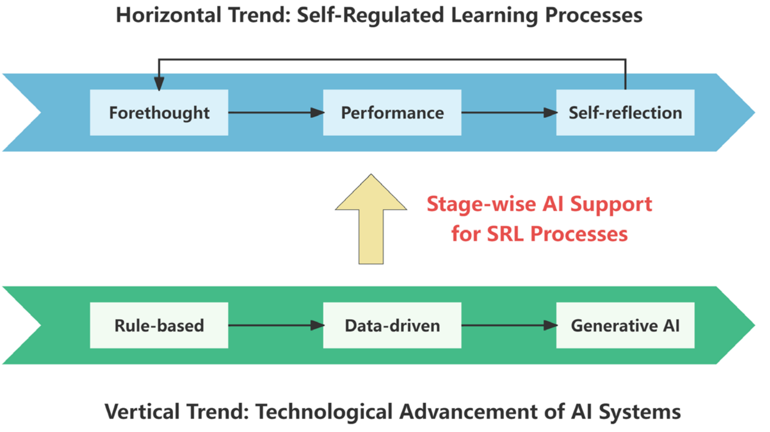

FIGURE 4 Conceptual framework: The intersection of AI evolution and SRL processes.

a range of inputs while placing the learner at the centre of the teaching and learning process. This enabled highly personalized, dynamically adaptive instruction (Liang et al., 2024).

Overall, the preliminary review suggests that AI support for SRL can be organized into three developmental paradigms. Figure 4 synthesizes this trajectory by mapping the technological evolution (vertical axis) against the cyclical nature of SRL processes (horizontal axis). Crucially, this vertical dimension represents a cumulative typology rather than a strictly chronological sequence, acknowledging that earlier rule-based approaches continue to function alongside emerging generative models. This framework facilitates a systematic examination of how the distinct affordances characterizing each paradigm intersect with specific phases of self-regulation.

Heterogeneity testing and effect size calculation (for RQ2)

Before calculating effect sizes, heterogeneity tests were performed to determine the most suitable meta-analysis model (Higgins & Thompson, 2002). Drawing on Zimmerman's SRL phases (forethought, performance, self-reflection and comprehensive evaluation), Cochran's Q-test and the I^2 statistic were used to examine potential heterogeneity. When the Q-test is significant ($p<0.05$) and I^2 exceeds 75%, the data are considered to show high heterogeneity (Higgins, 2011). As shown in Table 3, all subgroups yielded Q-test significance, with I^2 values above 75%. This considerable heterogeneity indicates that AI's impact on SRL is not monolithic, confirming the need for a random-effects model and subsequent moderator analysis. Consequently, a random-effects model was adopted to address the substantial variability introduced by different research settings, participant characteristics, measurement approaches and the diverse nature of control group conditions (e.g. lab-based tasks, authentic classroom settings), where the common element was the absence of the specific AI intervention.

Next, Hedges' g was employed to standardize effect sizes for AI interventions in each SRL phase (Hedges & Tipton, 2010). This statistic is widely used for synthesizing results across studies (Hedges, 2008). Calculations were carried out in Stata 18, using precise estimation methods and small-sample corrections (Hamza et al., 2008) to reduce biases stemming from small sample sizes and high heterogeneity. These findings indicate that interventions exert a clear influence during the performance phase but have a comparatively weaker impact in the preparatory (forethought) and evaluative (self-reflection) stages.

Based on Cohen's (2013) thresholds, where g values of 0.2, 0.5 and 0.8 denote small, medium and large effects, the forethought ($g=0.401$) and self-reflection ($g=0.464$) phases both showed small effect sizes. In contrast, the performance ($g=0.574$) and comprehensive

TABLE 3 Random-effects model for AI interventions in SRL phases.

	Effect size and 95% CI					Heterogeneity test				Tau-squared		
Variables	***k***	**Hedges' *g***	**SE**	**Lower**	**Upper**	***Q***	***df(Q)***	***p***	I^2	**Tau-squared**	***SE***	**Tau**
Forethought phase	33	0.401	0.123	0.160	0.643	271.43	32	0.0000	88.95	0.415	0.133	0.644
Performance phase	51	0.574	0.110	0.358	0.789	533.81	50	0.0000	92.05	0.474	0.119	0.688
Self-Reflection phase	18	0.464	0.196	0.079	0.848	194.45	17	0.0000	91.81	0.594	0.256	0.771
Comprehensive phase	31	0.529	0.103	0.328	0.730	204.14	30	0.0000	81.93	0.337	0.117	0.581
Overall	133	0.507	0.063	0.384	0.630	1208.24	132	0.0000	89.81	0.432	0.069	0.657

($g = 0.529$) phases reached medium levels. Collectively, the AI-supported interventions yielded a medium overall effect on SRL ($g = 0.507$, 95% CI [0.384, 0.630]). The results confirm a pattern of differential effectiveness across the regulatory cycle: interventions exert a clear influence during the performance phase but have a comparatively weaker impact in the preparatory (forethought) and evaluative (self-reflection) stages. These latter phases may demand more complex, internally driven cognitive and metacognitive processes (e.g. proactive goal setting, nuanced self-critique and causal attribution) that current AI designs are still developing capabilities to effectively foster, pointing to a need for improved support across the full regulatory cycle.

Sensitivity analysis and publication bias

To confirm the stability of these findings, 'leave-one-out' sensitivity tests were conducted. After sequentially excluding each study, the overall effect size remained consistent, ranging from 0.491 to 0.519. Subgroup analyses for individual SRL phases also displayed a similar degree of stability, suggesting that no single study unduly influenced the overall results.

Publication bias was examined through multiple approaches, including funnel plots, Egger's regression test, fail-safe N, and the trim-and-fill method.

While funnel plots for both the overall sample and individual phases appeared broadly symmetrical (Figures 5 and 6), Egger's regression tests revealed significant asymmetry for the overall sample ($z = 3.93$, $p = 0.0001$) and the performance phase ($z = 3.66$, $p = 0.0003$). This asymmetry is particularly concerning for the performance phase, where we observed the highest effect sizes ($g = 0.574$), suggesting that publication bias may be inflating our estimates precisely where AI appears most effective. The fail-safe N analysis indicated that 5908 null studies would be needed to invalidate the overall effect (exceeding Rosenthal's (1979) $5k + 10$ benchmark), with Orwin's (1983) calculation requiring 4451 studies to reduce the effect below 0.01.

The concentration of publication bias in the performance phase reflects a broader pattern in educational technology research, where interventions demonstrating immediate,

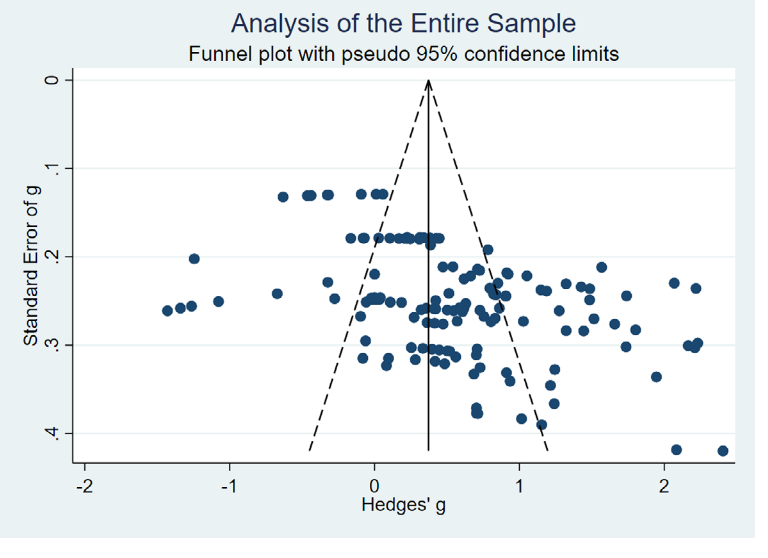

FIGURE 5 Funnel plot of overall sample.

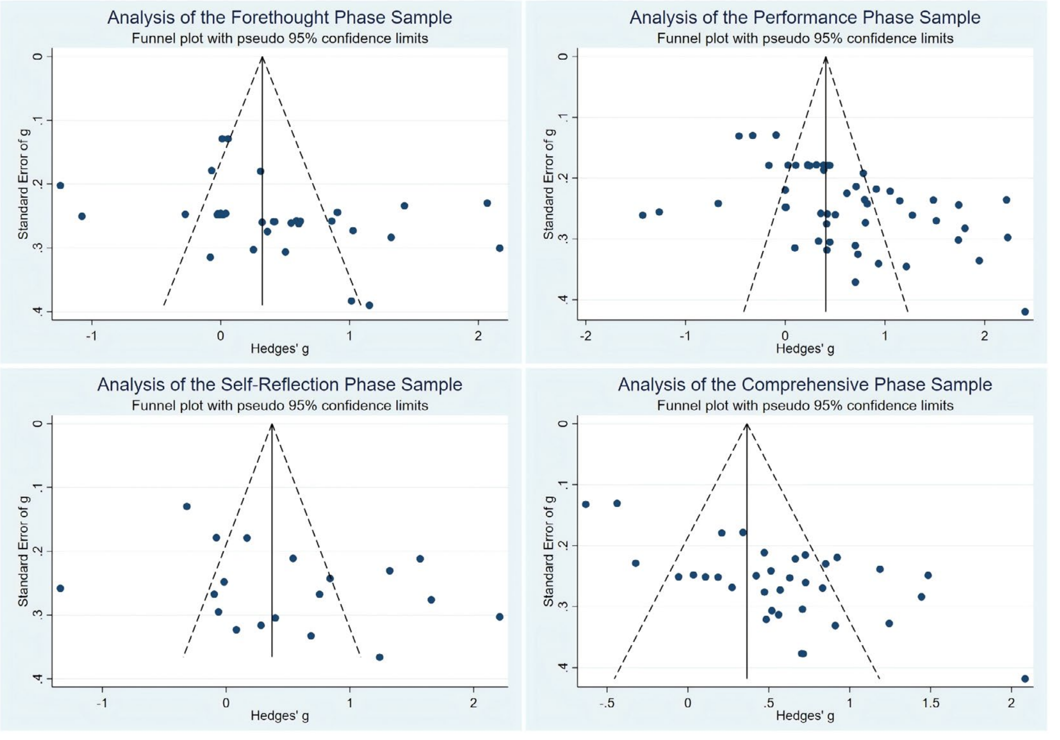

FIGURE 6 Funnel plots of the four SRL phases.

measurable gains are more likely to be published than those showing null or negative results (Cheung & Slavin, 2016). This selective reporting creates an overly optimistic picture of AI's capacity to enhance real-time learning performance, potentially misleading practitioners about the reliability of these interventions across diverse contexts.

The trim-and-fill method imputed three studies and adjusted the overall effect upwards from 0.507 to 0.569, suggesting minimal impact from missing studies. However, given the significant Egger's test results and the known limitations of trim-and-fill in distinguishing bias from heterogeneity (Peters et al., 2007), we interpret these effect sizes, particularly for the performance phase, as likely upper bounds rather than precise point estimates of AI's true impact. This interpretation aligns with current meta-analytic best practices (e.g. Wu & Yu, 2024) and underscores the need for systematic changes in publication practices, including pre-registration and commitment to publishing null results (Di Pietro & Castaño Muñoz, 2025), to build a more accurate evidence base for AI's role in supporting self-regulated learning (Table 4).

Subgroup analysis of AI technology evolution across SRL phases (for RQ3)

This study categorized the AI tools identified in the reviewed literature into three main developmental paradigms: rule-based systems, data-driven systems and GAI. To explore the significant heterogeneity observed in the initial analysis (I^2=89.81% overall), this section examines how intervention effectiveness varies as a function of the AI paradigm itself. Effect sizes were computed for both the overall dataset and for each phase of Zimmerman's SRL

TABLE 4 Measures of publication bias.

Self-regulated learning process	k	Egger's Score	Egger's SD	Egger's z	Egger's p	Fail-safe N Classic	Fail-safe N Orwin
Forethought phase	33	5.15	2.270	2.27	0.0232	594	930
Performance phase	51	5.94	1.622	3.66	0.0003	3167	1945
Self-reflection phase	18	4.08	3.333	1.22	0.2213	218	555
Comprehensive phase	31	5.86	3.384	1.73	0.0834	843	930
Overall	133	4.66	1.186	3.93	0.0001	5908	4451

Note: k, number of studies; Classic Fail-Safe N: Represents the number of null-effect studies needed to raise the p-value above the significance threshold ($\alpha=0.05$); Orwin's Fail-Safe N: Represents the number of missing studies required to reduce the effect size to a predefined trivial level (ES = 0.01).

TABLE 5 Combined effect sizes for AI developmental stages and SRL phases (Hedges' g).

SRL process	AI phase	k	Hedges' g	95% CI	I^2	p	Z
Forethought	Rule-based	2	–	–	–	–	–
	Data-driven	15	0.083	[−0.301, 0.466]	92.53%	0.6708	0.43
	Generative	16	0.709	[0.437, 0.981]	76.84%	0.0000	5.11
Performance	Rule-based	6	–	–	–	–	–
	Data-driven	32	0.352	[0.080, 0.624]	93.45%	0.0112	2.54
	Generative	13	0.938	[0.549, 1.328]	85.64%	0.0000	4.71
Self-reflection	Rule-based	2	–	–	–	–	–
	Data-driven	8	0.196	[−0.387, 0.779]	94.24%	0.5105	0.66
	Generative	8	0.826	[0.247, 1.405]	88.08%	0.0051	2.80
Comprehensive	Rule-based	13	0.429	[0.197, 0.662]	59.66%	0.0003	3.61
	Data-driven	13	0.580	[0.190, 0.971]	90.94%	0.0036	2.91
	Generative	8	0.682	[0.416, 0.938]	60.22%	0.0000	5.02
Overall	Rule-based	23	0.534	[0.339, 0.730]	66.77%	0.0000	5.34
	Data-driven	68	0.318	[0.134, 0.501]	93.07%	0.0006	3.41
	Generative	42	0.799	[0.604, 0.994]	82.34%	0.0000	8.03

Note: Caution is advised for subgroups with $k < 10$, while those with $k < 7$ may have limited reliability.

model (forethought, performance and self-reflection). Subgroups containing fewer than five studies were excluded due to limited interpretive value. Table 5 summarizes the combined effect sizes (Hedges' g) for each developmental stage across the SRL phases. The results show that overall effect sizes differ significantly across the three AI paradigms (Qb = 15.73, $p < 0.001$), progressing from rule-based systems ($g = 0.534$) and data-driven AI ($g = 0.318$) to GAI ($g = 0.799$).

Moderating effect analysis (for RQ4)

To identify potential sources of heterogeneity, this section reports the results of subgroup analyses on five study-level characteristics: educational level, discipline, learning space, duration and measurement type (Table 6). These analyses aim to reveal systematic patterns in intervention effectiveness and generate targeted hypotheses for future experimental validation.

TABLE 6 Subgroup analysis results of meta-analysis (Hedges' *g*).

Subgroup	*k*	*g*	95% CI	*Z* (*p*)	Between *p**
Educational level					0.221 (3.02)
Primary school	14	0.304	[0.060, 0.548]	2.44 (0.015)	
Junior and senior high school	11	0.636	[0.284, 0.989]	3.54 (0.001)	
Higher education	108	0.517	[0.374, 0.660]	7.09 (<0.001)	
Subject					<0.001 (21.09)
Natural Science	73	0.753	[0.576, 0.929]	8.36 (<0.001)	
Social Science	56	0.211	[0.061, 0.360]	2.77 (0.006)	
Combined subject areas	129	0.516	[0.390, 0.643]	8.00 (<0.001)	
Learning space					0.087 (2.93)
Online learning	37	0.732	[0.376, 1.087]	4.04 (<0.001)	
Blended learning	96	0.409	[0.309, 0.509]	8.02 (<0.001)	
Experimental duration					0.002 (12.44)
Less than 2 weeks	45	0.776	[0.551, 1.002]	6.88 (<0.001)	
2–10 weeks	43	0.361	[0.222, 0.500]	7.64 (<0.001)	
>10 weeks	45	0.374	[0.139, 0.609]	0.32 (<0.001)	
Overall	133	0.507	[0.384, 0.630]	8.08 (<0.001)	—

DISCUSSION

This meta-analysis reveals that while AI interventions yield a moderate overall benefit to self-regulated learning, their effectiveness is highly variable and contingent on the type of AI, the specific SRL phase, the learning context and the method of measurement. The following discussion interprets these findings through a critical theoretical lens, focusing on the distinction between supporting in-task performance and fostering internalized competence. This performance-competence divide emerges as a central theme for understanding the current state and future trajectory of AI in SRL research.

The overall effectiveness of AI in supporting SRL in context

The overall medium effect size ($g = 0.507$) confirms that AI-supported interventions are a valuable tool for enhancing SRL, with an impact comparable to other established educational interventions. Consistent with the medium effect size ($g = 0.50$) reported by Guo (2022) for metacognitive prompt interventions in SRL, the findings here suggest that AI-supported interventions, through the delivery of specific instructional features such as personalized feedback or scaffolded pathways during task execution (the performance phase), can serve functions analogous to metacognitive prompts in guiding certain aspects of learners' self-regulatory skills. It is important to recognize that the effectiveness likely stems from these carefully designed instructional strategies facilitated by AI, rather than AI acting as a monolithic causal agent. However, this study's overall effect size is lower than the value ($g = 0.897$) documented by D. Lee and Kwon (2024) in Korean K–12 classrooms—a discrepancy that may stem from SRL's long-term developmental trajectory or from variations in measurement approaches. This value is slightly higher than that reported by Dai et al. (2024) ($g = 0.40$, focusing on AI-driven virtual agents for learners' general competencies), highlighting the potential of appropriately designed AI-facilitated adaptive interventions to effectively support SRL, particularly when they are geared to foster personalized and self-directed learning processes.

The performance-competence divide: Interpreting effects across AI paradigms and SRL phases

The progression in overall effect sizes from rule-based systems ($g=0.534$) and data-driven AI ($g=0.318$) to GAI ($g=0.799$) is not merely a story of technological improvement. Instead, it must be interpreted through a critical theoretical lens that distinguishes between supporting in-task performance and fostering internalized regulatory competence. This distinction is particularly salient given that the synthesized effect sizes aggregate two distinct outcome families: learners' judgements about their SRL (via self-report) and their observable regulatory actions (via behavioural traces).

Because these measures capture different constructs and yield significantly different results, the apparent progression in overall effectiveness warrants a cautious interpretation regarding the enhancement of durable SRL competence. Rather than signalling definitive mastery, the data reflect a complex interplay between scaffolded performance and learners' perceptions of their capabilities, a tension we explore below.

Overall trends across AI paradigms

Rule-based AI: A foundational approach under continuous evolution

Although rule-based systems established a critical foundation ($g=0.534$) by providing stable scaffolding in structured environments (Duffy & Azevedo, 2015), their reliance on predefined prompts often limited their effectiveness in open-ended contexts (Chu et al., 2023). While early studies reported large effect sizes (Jones et al., 2018), later work revealed that fixed rule sets struggle to nurture transferable regulatory routines. Log-level evidence clarifies this mixed picture: although scripted prompts reliably triggered within-task planning and monitoring, subsequent segments of the same logfile showed a rapid decay of those behaviours once scaffolds disappeared, and the frequency of unprompted SRL actions rarely predicted posttest achievement (Bouchet et al., 2016; Long & Aleven, 2013). These findings suggest that while rule-based logic ensures clarity, its rigidity functions more as a procedural guide than a catalyst for adaptive self-regulation.

Data-driven AI: Advancing adaptive technology through dynamic analytics

Data-driven AI improved upon rule-based limitations by utilizing learning analytics to tailor interventions, yet its overall impact remains moderate and highly heterogeneous ($g\approx0.32$, $I^2\approx93\%$). For example, Duolingo showed notable success in supporting goal setting and performance monitoring (Qiao & Zhao, 2023; Wei, 2023). Such platforms typically achieve these outcomes by dynamically adapting curriculum content and task difficulty based on performance data (e.g. Weber et al., 2025), thereby influencing learning processes. At the same time, the effectiveness of data-driven systems can be closely tied to learners' self-regulation skills and comfort with technology. Some studies found that increased complexity lowered self-efficacy (Harati et al., 2021), especially during multidisciplinary or extended projects (Pereira & Díaz, 2021). Overall, data-driven AI excelled at capturing learning trajectories and offering immediate feedback (Dever et al., 2023), such as automated corrective feedback in language learning (Liu et al., 2025).

The divergence in outcomes likely stems from the gap between information availability and pedagogical mediation. While real-time dashboards heighten situational awareness, they may also overwhelm novices when analytics are surfaced without interpretive support (Yang et al., 2024). This divergence appears rooted in socio-emotional and cognitive responses, as novice learners facing high cognitive loads may hesitate to seek help due to feelings of embarrassment, a barrier that well-designed, on-demand AI scaffolding can effectively mitigate (Huang et al., 2025). Although survey data often register feelings of personalization, trace studies indicate that these perceptions do not reliably translate into durable regulatory routines (Cabı & Türkoğlu, 2025). This suggests sophisticated monitoring can amplify SRL only when paired with scaffolds that help learners interpret—and act on—algorithmic signals.

Generative AI: Personalized companionship and multimodal integration

Generative AI achieved superior outcomes ($g=0.799$) by functioning as a responsive learning companion rather than a mere delivery system, though high heterogeneity ($I^2=82.34\%$) indicates that efficacy depends heavily on instructional design. Unlike earlier paradigms, GAI's capacity for highly adaptive, dialogic interaction serves to motivate learners and scaffold complex problem-solving (Pan et al., 2025; Wang et al., 2024). However, this very sophistication introduces the risk of fostering a polished dependency. Recent behavioural evidence cautions that intrusive or overly helpful support may foster ‘simulated SRL’ rather than durable competence, as learners' prompted elaborations can vanish once the tutor is withdrawn (Huang et al., 2025). This aligns with findings that overreliance on LLM-based feedback may even hinder performance when the support is unavailable, suggesting that short-term gains can mask a failure to internalize regulatory skills (Sun et al., 2025).

This discrepancy highlights a critical performance-competence divide. Current research reliance on immediate, single-wave measures leaves the durability of reported gains largely untested (Guan et al., 2025). This challenge is particularly acute for far transfer, defined as the application of metacognitive skills to structurally dissimilar tasks requiring different cognitive strategies (Schuster et al., 2020), which is the true hallmark of internalized competence. The available evidence, which rarely assesses far transfer, tends to focus on near transfer, leaving AI's capacity to foster robust, generalizable regulation an open and critical question. Therefore, while GAI demonstrates significant potential, the high variance suggests its effectiveness is not intrinsic to the technology but is contingent on designs that mitigate dependency. Without robust evidence of skill transfer, the technical richness that enables personalized companionship risks eroding learner autonomy. To harness GAI's strengths, future research must move beyond measuring immediate performance and adopt longitudinal designs that triangulate trace data, delayed achievement, and qualitative evidence. Furthermore, AI systems should be engineered with adaptive mechanisms, such as the guidance-based approaches that provide hints over direct answers (Lee et al., 2024), or scaffolds that strategically fade over time.

Phase-specific effects across AI paradigms

The substantial within-phase heterogeneity confirms that effectiveness depends on how well specific AI affordances address the distinct cognitive bottlenecks of each SRL phase.

In the forethought phase, GAI's marked advantage ($g=0.709$) over data-driven AI ($g=0.083$) reflects its unique capacity to operationalize abstract intentions. GAI systems excel in goal-setting because they force learners to externalize and refine their objectives through dialogue (Pan et al., 2025), whereas data-driven systems typically support only the review of past performance without actively scaffolding future planning (Yang et al., 2024). The limited rule-based AI evidence ($k=2$) suggests these systems can prompt basic planning behaviours but lack the flexibility to address the nuances of motivational beliefs.

During the performance phase, differential effects reveal how technology aligns with the demand for real-time adaptation. GAI's superior effect ($g=0.938$) indicates that its co-regulatory capabilities effectively reduce cognitive load during complex problem-solving (Ng et al., 2024). Data-driven AI's moderate but variable effect ($g=0.352$) reflects its strength in self-monitoring via dashboards but reveals a limitation in offering the immediate, corrective guidance required when learners encounter obstacles (Lim et al., 2023). Rule-based systems show promise here for structured strategy training through explicit metacognitive prompts (Dever et al., 2023), though their rigidity limits their utility in dynamic tasks.

In the self-reflection phase, the pattern illuminates a critical distinction between information availability and meaning-making. GAI's effectiveness ($g=0.826$) derives from scaffolding the interpretive process of self-evaluation and causal attribution (Lee et al., 2024). Conversely, data-driven AI's minimal impact ($g=0.196$) reveals that presenting performance metrics alone is insufficient; learners require support to translate raw data into adaptive inferences for future learning (Mejeh et al., 2024). This disparity underscores that effective support for self-reflection requires dialogic guidance that helps learners internalize the 'why' behind their performance. Ultimately, these analyses reveal that AI effectiveness depends not on technological sophistication per se, but on the alignment between specific AI affordances and particular self-regulatory processes.

Contextual and methodological moderators of AI effectiveness

Our moderator analysis reveals how study-level characteristics shape the effectiveness of AI interventions. As established in the results section, this analysis must be interpreted with considerable caution because these demographic-like variables represent distal factors and broad categories conceal significant internal heterogeneity. Thus, the purpose of this discussion is not to establish definitive causal pathways but to explore systematic patterns that signal underlying mechanisms and generate targeted hypotheses for future experimental validation.

Educational stage

The analysis revealed varying effectiveness across educational stages, with AI interventions yielding a small effect for primary school students ($g=0.304$), a larger effect for secondary students ($g=0.636$), and a moderate effect for higher education learners ($g=0.517$). This progression reflects developmental differences in metacognitive readiness and capacity for processing AI-generated feedback. Secondary students, at a critical juncture in developing self-regulatory skills, appear particularly responsive to the structured scaffolding AI provides (Ng et al., 2024). The pronounced effect at this level may also indicate that secondary students face greater challenges in SRL tasks without AI support, making the differential impact of AI interventions more apparent. Primary students' limited response stems from their developing cognitive maturity, which constrains their ability to independently utilize AI feedback without substantial teacher mediation (Liu, Hwang, et al., 2024). Higher education students, while possessing advanced SRL skills, still derive moderate benefits from AI support in managing complex cognitive tasks (Zhang, 2024).

Disciplinary domain

Natural science interventions demonstrated substantially larger effects ($g=0.753$) compared to social science interventions ($g=0.211$), with the latter approaching the threshold of small effect size. This marked disparity reflects the alignment between current AI capabilities and disciplinary task characteristics. Natural science problems typically feature structured procedures and clear solution paths, enabling AI to provide targeted diagnostic feedback and procedural scaffolding (Dever et al., 2023). In contrast, social science tasks involve open-ended inquiry, critical analysis of multiple perspectives and synthesis of ambiguous information—cognitive processes that challenge AI's capacity to offer nuanced support for dialectical reasoning and higher order thinking (Zhang & Hew, 2024). This suggests that effective AI scaffolding must be calibrated not only to content but also to the epistemological demands and ways of knowing inherent to different disciplines.

Learning space

Fully online environments yielded larger effects ($g=0.732$) than blended settings ($g=0.409$). This advantage stems from continuous AI tool availability in online contexts, enabling consistent engagement with regulatory support while minimizing external interferences (Zhang, 2024). However, this finding must be interpreted against the backdrop that online learning environments typically present heightened SRL challenges, potentially amplifying the relative benefit of AI support. Blended environments face the additional complexity of integrating AI-mediated and teacher-led instruction, where misalignment between online and offline components can diminish overall effectiveness (Zhang et al., 2023). This challenge is amplified when AI-driven feedback conflicts with teacher guidance, or when its autonomous nature clashes with the pedagogical norms of examination-oriented cultures, which may predispose students to specific, teacher-validated learning strategies (Pan et al., 2025). Optimal implementation requires careful orchestration of both modalities to create coherent learning experiences that leverage the strengths of each approach (Hsu et al., 2023).

Intervention duration

The counter-intuitive finding that brief interventions (<2 weeks) produced the largest effects ($g=0.776$) compared to medium (2–10 weeks, $g=0.361$) and extended durations (>10 weeks, $g=0.374$) aligns with patterns observed in some technology-enhanced learning research. This phenomenon likely reflects what Hooshyar et al. (2024) identify as novelty effects, where initial exposure to AI support generates heightened engagement and immediate performance gains that may not translate into durable SRL skill development. Medium-duration interventions provide a more realistic window for meaningful skill acquisition, offering adequate practice opportunities while maintaining engagement (Liu, Xu, et al., 2024). Extended interventions beyond 10 weeks may encounter diminishing returns due to technology fatigue, over-reliance on external scaffolding or what Buçinca et al. (2021) term 'cognitive forcing functions'—where prolonged AI support inadvertently undermines the development of autonomous regulation. This temporal pattern underscores that effective AI support for SRL requires not merely duration but carefully designed scaffolding that progressively fades to promote internalization of regulatory strategies.

Measurement type

A critical finding emerged from the analysis of measurement type: effects differed significantly by outcome family. Studies using behavioural/event traces showed a large effect ($g = 0.751$), more than double the small-to-moderate effect from self-reports ($g = 0.369$). Prior meta-analyses noted similar numerical trends but lacked sufficient behavioural studies to reach significance (Zhao et al., 2025). Our sample provides the statistical power to establish this difference. We attribute the larger behavioural effect to AI's capacity to orchestrate observable in-task regulation, which is captured in rich trace data (Winne, 2022). By contrast, the attenuated self-report effect likely reflects two complementary mechanisms. First, learners' conservative self-reports may reflect an accurate awareness that while performance improved, durable (Mejeh et al., 2024; Zhang, 2024), autonomous competence has not yet consolidated, a conclusion supported by evidence that AI scaffolds can yield 'simulated SRL' that decays once removed (Huang et al., 2025) or foster overreliance (Sun et al., 2025). Second, from a methodological standpoint, survey instruments themselves are reactive; the process of completing them can prompt metacognitive reflection across all conditions, thereby diluting the measured treatment effect. Collectively, this discrepancy, when considered alongside the field's general lack of long-term follow-up data, suggests that current tools may be more effective at boosting immediate, tool-dependent performance than at cultivating durable, transferable competence.

Conclusion and implications

This meta-analysis synthesized a decade of research on AI-supported self-regulated learning and revealed a fundamental tension: artificial intelligence demonstrates clear potential to enhance self-regulated learning. However, this potential is not a guaranteed outcome of technological sophistication. Instead, it is unevenly realized across different phases of the regulatory cycle and remains heavily contingent on specific technological, pedagogical and contextual factors. Beyond mere technological advancement, the observed evolutionary trajectory—from rule-based systems through data-driven approaches to generative AI—signifies a paradigm shift in conceptualizing technology's role in supporting learner autonomy. This progression suggests that effective AI support for self-regulated learning requires not merely sophisticated algorithms but careful alignment between technological affordances and the cognitive-motivational processes underlying self-regulation. The high heterogeneity observed even within the most advanced AI paradigms underscores that efficacy is less about the tool itself and more about the pedagogical precision of its implementation. Building on these conclusions, the following implications for future research and practice are proposed to address the field's most pressing challenges.

The future of traditional AI: From silos to smart integration

Many educational settings still employ rule-based and data-driven AI systems (Sarker et al., 2024). While these systems are less effective than GAI in supporting motivational and reflective processes (Pelánek et al., 2024), future research should explore intelligent integration strategies. A hybrid approach where rule-based systems provide stable structural scaffolding while GAI adds personalized, emotionally responsive elements could leverage the predictability of traditional AI as a foundation upon which more sophisticated interactions are built.

The paradox of generative AI: Balancing empowerment and dependence

Although GAI shows the highest pooled outcomes, the significant variance suggests that its human-like responsiveness can be a double-edged sword: it may maximally support performance or inadvertently deprive learners of necessary cognitive struggles that foster genuine self-regulation (Lin & Yu, 2025). The scaffolding that makes GAI most supportive may also create insidious forms of intellectual dependence. Future work must develop 'productive failure' mechanisms within GAI systems—strategic moments where support is deliberately withdrawn to promote autonomous problem-solving—and investigate designs that feature strategically fading scaffolds over time.

Context is key: Adapting AI to diverse educational ecologies

While our moderator analysis reveals significant variations across educational stages and disciplines, the effectiveness of AI-supported SRL also depends on unexplored sociocultural dynamics. Future research must move beyond treating context as demographic variables to examining the relational and cultural dimensions—such as the interplay between teacher guidance and AI feedback or cross-cultural responses to AI's emphasis on autonomy—that fundamentally determine whether AI scaffolding translates into meaningful learning support.

From supporting performance to fostering competence: Bridging the core divide

This synthesis culminates in a fundamental challenge for the field: bridging the gap between enhancing immediate, scaffolded task performance and developing transferable, internalized self-regulatory competence. This distinction, highlighted by the differential effects across SRL phases and measurement types, emerges as our most pressing research imperative. Future assessment frameworks must incorporate delayed measurements in unsupported environments and longitudinal tracking, shifting focus from 'Does AI improve learning outcomes?' to 'Does AI cultivate learners who can regulate their learning independently?'

LIMITATIONS AND SUGGESTIONS

Although this study made noteworthy progress by systematically reviewing and quantitatively synthesizing research on AI technologies that support SRL, its conclusions must be interpreted in light of several limitations. First, the findings are shaped by the methodological characteristics of the primary research synthesized. Many included studies were conducted in controlled laboratory settings over brief periods. This context, combined with our analytical aggregation of diverse outcome measures at the phase level, means that the reported effect sizes represent a high-level synthesis whose generalizability to real-world implementation requires careful consideration. This challenge is compounded by the considerable variability in how the primary studies operationalized transfer, which often remained undifferentiated between near- and far-transfer tasks—a critical distinction for evaluating authentic competence.

Furthermore, our analysis reveals a fundamental measurement heterogeneity that constrains interpretation. The predominance of self-report measures introduces potential reactivity, as survey instruments may function as a metacognitive intervention for all participants, thus potentially underestimating the AI's true effect. Conversely, while the significantly larger effects found in behavioural data confirm AI's capacity to scaffold in-task performance, this pattern simultaneously suggests that the observed gains may reflect a scaffold-dependent performance that dissipates upon tool withdrawal. This tension between different measurement modalities underscores the central interpretive challenge in bridging the performance-competence divide.

Second, beyond the constraints inherited from the primary studies, a further limitation lies in the nature of the moderators themselves analysed in this synthesis. Broad categories such as 'disciplinary domain' mask substantial internal heterogeneity, which constrains the precision of our findings and precludes strong causal claims. Future inquiry should therefore shift focus from whether AI is effective in broad contexts to how specific AI affordances mediate the relationship between scaffolded performance and internalized competence. This requires investigating more proximal, process-oriented variables. For instance, future meta-analyses could code for the interplay between specific AI-driven scaffolds (e.g. the fading of prompts, the nature of dialogic feedback) and fine-grained learner actions captured in trace data (e.g. strategy revision, metacognitive monitoring). Such an approach would yield more targeted, actionable insights into designing AI systems that effectively bridge the performance-competence divide.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflict of interest with respect to the research, authorship and publication of this article.

DATA AVAILABILITY STATEMENT

All data presented in this table were collected by the author. The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS APPROVAL STATEMENT

This study does not involve human or animal subjects.

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APPENDIX A

CHARACTERISTICS OF INCLUDED STUDIES

ID	Study (author and year)	Publication type	Title	*N*	AI paradigm	MERSQI
1	Bouchet, F., et al. (2013)	Conference	Impact of Different Pedagogical Agents' Adaptive Self-regulated Prompting Strategies on Learning with MetaTutor	40	Rule-based	12
2	Bouchet, F., et al. (2016)	Conference	Can adaptive pedagogical agents' prompting strategies improve students' learning and self-regulation?	58	Rule-based	14
3	Cabı, E., & Türkoğlu, H. (2025)	Journal	The Impact of a Learning Analytics Based Feedback System on Students' Academic Achievement and Self-Regulated Learning in a Flipped Classroom	127	Data-driven	13.5
4	Chang, C.-Y., et al. (2024)	Journal	Promoting students' learning achievement and self-efficacy: A mobile chatbot approach for nursing training	36	Data-driven	12.5
5	Chu, S.-T., et al. (2023)	Journal	Incorporating teacher intelligence into digital games: An expert system-guided self-regulated learning approach…	44	Rule-based	12.5
6	Dever, D. A., et al. (2023)	Journal	A complex systems approach to analysing pedagogical agents' scaffolding of self-regulated learning within an intelligent tutoring system	117	Data-driven	13
7	Duffy, M. C., & Azevedo, R. (2015)	Journal	Motivation matters: Interactions between achievement goals and agent scaffolding for self-regulated learning within an intelligent tutoring system	83	Rule-based	12.5
8	Fleur, D. S., et al. (2023)	Journal	IguideME: Supporting Self-Regulated Learning and Academic Achievement with Personalized Peer-Comparison Feedback in Higher Education	41	Data-driven	14.5
9	Harati, H., et al. (2021)	Journal	Assessment and Learning in Knowledge Spaces (ALEKS) Adaptive System Impact on Students' Perception and Self-Regulated Learning Skills	240	Data-driven	10
10	Hilpert, J. C., et al. (2023)	Journal	Leveraging complexity frameworks to refine theories of engagement: Advancing self-regulated learning in the age of artificial intelligence	143	Data-driven	14
11	Hsu, T.-C. (2023)	Journal	Artificial Intelligence image recognition using self-regulation learning strategies: effects on vocabulary acquisition…	47	Generative AI	13

ID	Study (author and year)	Publication type	Title	*N*	AI paradigm	MERSQI
12	Huang, A. Y. Q., et al. (2025)	Journal	The impact of GenAI-enabled coding hints on students' programming performance and cognitive load in an SRL-based Python course	31	Generative AI	14
13	Jones, A., et al. (2018)	Conference	'I Know That Now, I'm Going to Learn This Next' Promoting Self-regulated Learning with a Robotic Tutor	45	Rule-based	13
14	Lee, H.-Y., et al. (2024)	Journal	Empowering ChatGPT with guidance mechanism in blended learning: effect of self-regulated learning, higher-order thinking skills…	61	Generative AI	14
15	Liang, H.-Y., et al. (2024)	Journal	Effect of an AI-based chatbot on students' learning performance in alternate reality game-based museum learning	39	Generative AI	13.5
16	Liao, X., et al. (2024)	Journal	Design and implementation of an AI-enabled visual report tool as formative assessment to promote learning achievement…	125	Data-driven	12.5
17	Lim, L., et al. (2023)	Journal	Effects of real-time analytics-based personalized scaffolds on students' self-regulated learning	64	Rule-based	13.5
18	Liu, C.-C., et al. (2025)	Journal	Effects of an automated corrective feedback-based peer assessment approach on students' learning achievement, motivation, and self-regulated learning…	66	Data-driven	13.5
19	Liu, Z.-M., et al. (2024)	Journal	Integrating large language models into EFL writing instruction: effects on performance, self-regulated learning strategies, and motivation	65	Generative AI	14
20	Long, Y., & Aleven, V. (2013)	Conference	Supporting Students' Self-Regulated Learning with an Open Learner Model in a Linear Equation Tutor	56	Rule-based	12.5
21	Mejeh, M., et al. (2023)	Journal	Effects of adaptive feedback through a digital tool – a mixed-methods study on the course of self-regulated learning	66	Data-driven	12.5
22	Ng, D. T. K., et al. (2024)	Journal	Empowering student self-regulated learning and science education through ChatGPT: A pioneering pilot study	74	Generative AI	12.5
23	Pan, M., et al. (2025)	Journal	Effects of GenAI-empowered interactive support on university EFL students' self-regulated strategy use and engagement in reading	61	Generative AI	14
24	Pereira, J., & Díaz, Ó. (2021)	Journal	Struggling to Keep Tabs on Capstone Projects: A Chatbot to Tackle Student Procrastination	53	Data-driven	11

ID	Study (author and year)	Publication type	Title	*N*	AI paradigm	MERSQI
25	Qiao, H., & Zhao, A. (2023)	Journal	Artificial intelligence-based language learning: illuminating the impact on speaking skills and self-regulation in Chinese EFL context	93	Data-driven	15
26	Sarkhosh, M. (2018)	Journal	The Effects of Self-Regulatory Learning through Computer-Assisted Intelligent Tutoring System on the Improvement of EFL Learner' Speaking Ability	30	Rule-based	10.5
27	Su, J. M. (2014)	Conference	A Self-Regulated Learning System to Support Adaptive Scaffolding in Hypermedia-based Learning Environments	86	Rule-based	13
28	Sun, D., et al. (2025)	Journal	How Self-Regulated Learning Is Affected by Feedback Based on Large Language Models: Data-Driven Sustainable Development…	63	Generative AI	12.5
29	Wang, W.-S., et al. (2024)	Journal	Signaling feedback mechanisms to promoting self-regulated learning and motivation in virtual reality transferred to real-world…	71	Generative AI	15
30	Weber, F., et al. (2025)	Journal	Enhancing legal writing skills: The impact of formative feedback in a hybrid intelligence learning environment	43	Data-driven	14.5
31	Wei, L. (2024)	Journal	Artificial intelligence in language instruction: impact on English learning achievement, L2 motivation, and self-regulated learning	60	Data-driven	13.5
32	Wirzberger, M., et al. (2023)	Journal	Optimal feedback improves behavioral focus during self-regulated computer-based work	127	Data-driven	13.5
33	Yang, C. C. Y., et al. (2024)	Journal	Learning analytics dashboard-based self-regulated learning approach for enhancing students' e-book-based blended learning	83	Data-driven	13
34	Zhang, L., & Hew, K. F. (2024)	Journal	Leveraging unlabeled data: Fostering self-regulated learning in online education with semi-supervised recommender systems	84	Data-driven	10
35	Zhang, M. (2024)	Journal	Enhancing self-regulation and learner engagement in L2 speaking: exploring the potential of intelligent personal assistants…	54	Data-driven	13.5